0

MIG Welding Station
No. 1

MIG_Interface	
MIG_Interface	Welder1 ...
Arc_Start	Local:3:I.Data.0 0
Arc_End	Local:3:I.Data.1 0
Wire_Feed	Local:4:O.Data.0 0
Gas_Flow	Local:4:O.Data.1 0
Arc_On	Local:4:O.Data.2 0
Pre_Weld_Delay	Welder1.Pre_Weld_Delay 10000
Fault_Arc	Local:3:I.Data.2 0
Fault_Wire	Local:3:I.Data.3 0

1

MIG Welding Station
Nno. 2

MIG_Interface	
MIG_Interface	Welder2 ...
Arc_Start	Local:3:I.Data.4 0
Arc_End	Local:3:I.Data.5 0
Wire_Feed	Local:4:O.Data.3 0
Gas_Flow	Local:4:O.Data.4 0
Arc_On	Local:4:O.Data.5 0
Pre_Weld_Delay	Welder2.Pre_Weld_Delay 0
Fault_Arc	Local:3:I.Data.6 0
Fault_Wire	Local:3:I.Data.7 0

(End)

Name	Value	Data Type	Scope
Local:3:I		AB:1756_DI:I:0	Add_On_MIG
Constant	No		
External Access:	Read/Write		
OPC UA Access:	None		
Local:3:I.Data.0	0	BOOL	
Local:3:I.Data.0 - MainProgram/MainRoutine - 0(MIG_Interface)			
*Local:3:I.Data.0 - Model/Emulation - *1(OTE)*			
Local:3:I.Data.1	0	BOOL	
Local:3:I.Data.1 - MainProgram/MainRoutine - 0(MIG_Interface)			
*Local:3:I.Data.1 - Model/Emulation - *2(OTE)*			
Local:3:I.Data.2	0	BOOL	
Local:3:I.Data.2 - MainProgram/MainRoutine - 0(MIG_Interface)			
*Local:3:I.Data.2 - Model/Emulation - *3(OTE)*			
Local:3:I.Data.3	0	BOOL	
Local:3:I.Data.3 - MainProgram/MainRoutine - 0(MIG_Interface)			
*Local:3:I.Data.3 - Model/Emulation - *4(OTE)*			
Local:3:I.Data.4	0	BOOL	
Local:3:I.Data.4 - MainProgram/MainRoutine - 1(MIG_Interface)			
*Local:3:I.Data.4 - Model/Emulation - *5(OTE)*			
Local:3:I.Data.5	0	BOOL	
Local:3:I.Data.5 - MainProgram/MainRoutine - 1(MIG_Interface)			
*Local:3:I.Data.5 - Model/Emulation - *6(OTE)*			
Local:3:I.Data.6	0	BOOL	
Local:3:I.Data.6 - MainProgram/MainRoutine - 1(MIG_Interface)			
*Local:3:I.Data.6 - Model/Emulation - *7(OTE)*			
Local:3:I.Data.7	0	BOOL	
Local:3:I.Data.7 - MainProgram/MainRoutine - 1(MIG_Interface)			
*Local:3:I.Data.7 - Model/Emulation - *8(OTE)*			
Local:4:O		AB:1756_DO:O:0	Add_On_MIG
Constant	No		
External Access:	Read/Write		
OPC UA Access:	None		
Local:4:O.Data.0	0	BOOL	
*Local:4:O.Data.0 - MainProgram/MainRoutine - *0(MIG_Interface)*			
Local:4:O.Data.1	0	BOOL	
*Local:4:O.Data.1 - MainProgram/MainRoutine - *0(MIG_Interface)*			
Local:4:O.Data.2	0	BOOL	
*Local:4:O.Data.2 - MainProgram/MainRoutine - *0(MIG_Interface)*			
Local:4:O.Data.3	0	BOOL	
*Local:4:O.Data.3 - MainProgram/MainRoutine - *1(MIG_Interface)*			
Local:4:O.Data.4	0	BOOL	
*Local:4:O.Data.4 - MainProgram/MainRoutine - *1(MIG_Interface)*			
Local:4:O.Data.5	0	BOOL	
*Local:4:O.Data.5 - MainProgram/MainRoutine - *1(MIG_Interface)*			
Welder1		MIG_Interface	MainProgram
MIG Welding Station No. 1			
Constant	No		
External Access:	Read/Write		
OPC UA Access:	None		
*Welder1 - MainProgram/MainRoutine - *0(MIG_Interface)*			
Welder1.EnableIn	1	BOOL	
MIG Welding Station No. 1 Enable Input - System Defined Parameter			
Welder1.EnableOut	0	BOOL	
MIG Welding Station No. 1 Enable Output - System Defined Parameter			
Welder1.Arc_Start	0	BOOL	
MIG Welding Station No. 1 Begin MIG Welding			
Welder1.Arc_End	0	BOOL	
MIG Welding Station No. 1 MIG Welding Finished			
Welder1.Wire_Feed	0	BOOL	
MIG Welding Station No. 1 Energize Filler Wire Feeder			
Welder1.Gas_Flow	0	BOOL	

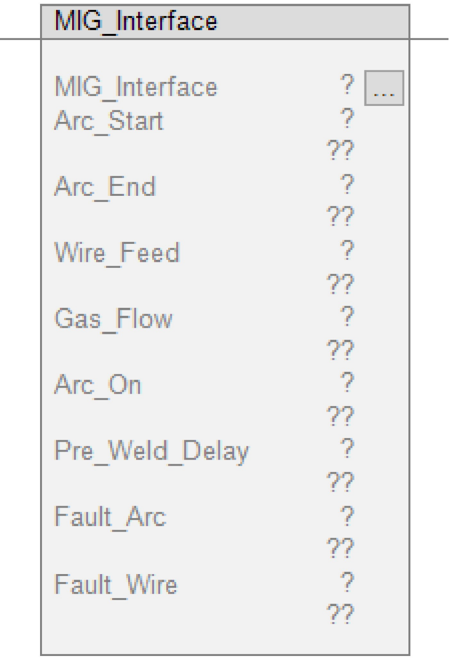
Welder1 (Continued)
MIG Welding Station No. 1 Open Shielding Gas Valve
Welder1.Arc_On 0 BOOL
MIG Welding Station No. 1 MIG Arc Established
Welder1.Pre_Weld_Delay 10000 DINT
MIG Welding Station No. 1 Gas and Wire Preflow
Welder1.Pre_Weld_Delay - MainProgram/MainRoutine - 0(MIG_Interface)
Welder1.Fault_Arc 0 BOOL
MIG Welding Station No. 1 Loss of MIG Welding Arc
Welder1.Fault_Wire 0 BOOL
MIG Welding Station No. 1 Wire feed not detected

Welder2 MIG_Interface MainProgram
MIG Welding Station Nno. 2
Constant No
External Access: Read/Write
OPC UA Access: None
*Welder2 - MainProgram/MainRoutine - *1(MIG_Interface)*
Welder2.EnableIn 1 BOOL
MIG Welding Station Nno. 2 Enable Input - System Defined Parameter
Welder2.EnableOut 0 BOOL
MIG Welding Station Nno. 2 Enable Output - System Defined Parameter
Welder2.Arc_Start 0 BOOL
MIG Welding Station Nno. 2 Begin MIG Welding
Welder2.Arc_End 0 BOOL
MIG Welding Station Nno. 2 MIG Welding Finished
Welder2.Wire_Feed 0 BOOL
MIG Welding Station Nno. 2 Energize Filler Wire Feeder
Welder2.Gas_Flow 0 BOOL
MIG Welding Station Nno. 2 Open Shielding Gas Valve
Welder2.Arc_On 0 BOOL
MIG Welding Station Nno. 2 MIG Arc Established
Welder2.Pre_Weld_Delay 0 DINT
MIG Welding Station Nno. 2 Gas and Wire Preflow
Welder2.Pre_Weld_Delay - MainProgram/MainRoutine - 1(MIG_Interface)
Welder2.Fault_Arc 0 BOOL
MIG Welding Station Nno. 2 Loss of MIG Welding Arc
Welder2.Fault_Wire 0 BOOL
MIG Welding Station Nno. 2 Wire feed not detected

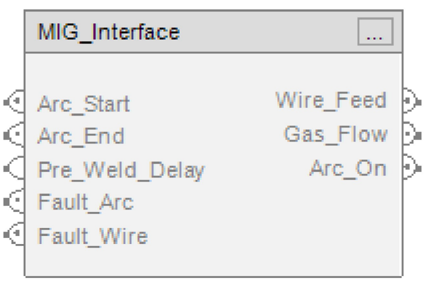
MIG_Interface v1.0

Available Languages

Relay Ladder



Function Block



Structured Text

MIG_Interface(Arc_Start, Arc_End, Wire_Feed, Gas_Flow, Arc_On, Pre_Weld_Delay, Fault_Arc, Fault_Wire);

Parameters

Required	Name	Data Type	Usage	Description
X	MIG_Interface	MIG_Interface	InOut	
	EnableIn	BOOL	Input	
	EnableOut	BOOL	Output	
X	Arc_Start	BOOL	Input	Begin MIG Welding
X	Arc_End	BOOL	Input	MIG Welding Finished
X	Wire_Feed	BOOL	Output	Energize Filler Wire Feeder
X	Gas_Flow	BOOL	Output	Open Shielding Gas Valve
X	Arc_On	BOOL	Output	MIG Arc Established
X	Pre_Weld_Delay	DINT	Input	Gas and Wire Preflow
X	Fault_Arc	BOOL	Input	Loss of MIG Welding Arc
X	Fault_Wire	BOOL	Input	Wire feed not detected

Extended Description

Execution

Condition	Description
EnableIn is true	

Revision v1.0 Notes

Name	Default	Data Type	Scope
Arc_End	0	BOOL	MIG_Interface

MIG Welding Finished
Usage: Input Parameter
Required: Yes
Visible: Yes
External Access: Read/Write
OPC UA Access: None
Arc_End - MIG_Interface/Logic - 1-A2(BNOT,BNOT_01.In), 1-A2(IREF,Arc_End)

Name	Default	Data Type	Scope
Arc_On	0	BOOL	MIG_Interface

MIG Arc Established
Usage: Output Parameter
Required: Yes
Visible: Yes
External Access: Read Only
OPC UA Access: None
*Arc_On - MIG_Interface/Logic - *1-E2(OREF,Arc_On), 1-C1(TONR,TONR_01.DN)*

Name	Default	Data Type	Scope
Arc_Start	0	BOOL	MIG_Interface

Begin MIG Welding
Usage: Input Parameter
Required: Yes
Visible: Yes
External Access: Read/Write
OPC UA Access: None
Arc_Start - MIG_Interface/Logic - 1-A1(IREF,Arc_Start), 1-B1(BOR,BOR_01.In1)

Name	Default	Data Type	Scope
Fault_Arc	0	BOOL	MIG_Interface

Loss of MIG Welding Arc
Usage: Input Parameter
Required: Yes
Visible: Yes
External Access: Read/Write
OPC UA Access: None
Fault_Arc - MIG_Interface/Logic - 1-A2(IREF,Fault_Arc), 1-C1(BAND,BAND_01.In3)

Name	Default	Data Type	Scope
Fault_Wire	0	BOOL	MIG_Interface

Wire feed not detected
Usage: Input Parameter
Required: Yes
Visible: Yes
External Access: Read/Write
OPC UA Access: None
Fault_Wire - MIG_Interface/Logic - 1-A2(IREF,Fault_Wire), 1-C1(BAND,BAND_01.In2)

Name	Default	Data Type	Scope
Gas_Flow	0	BOOL	MIG_Interface

Open Shielding Gas Valve
Usage: Output Parameter
Required: Yes
Visible: Yes
External Access: Read Only
OPC UA Access: None
*Gas_Flow - MIG_Interface/Logic - *1-E1(OREF,Gas_Flow), 1-C1(TONR,TONR_01.EN)*

Name	Default	Data Type	Scope
Pre_Weld_Delay	0	DINT	MIG_Interface

Gas and Wire Preflow
Usage: Input Parameter
Required: Yes
Visible: Yes
External Access: Read/Write
OPC UA Access: None
Pre_Weld_Delay - MIG_Interface/Logic - 1-C1(TONR,TONR_01.PRE), 1-C2(IREF,Pre_Weld_Delay)

Wire_Feed	0	BOOL	MIG_Interface

Energize Filler Wire Feeder

Usage:	Output Parameter
Required:	Yes
Visible:	Yes
External Access:	None
OPC UA Access:	None

*Wire_Feed - MIG_Interface/Logic - *1-E1(OREF,Wire_Feed), 1-C1(TONR,TONR_01.EN)*

Name	Default	Data Type	Scope
BAND_01		FBD_BOOLEAN_AND	MIG_Interface
Usage:	Local Tag		
External Access:	None		
OPC UA Access:	None		

*BAND_01 - MIG_Interface/Logic - *1-A2(BNOT,BNOT_01.Out), *1-A2(IREF,Fault_Arc), *1-A2(IREF,Fault_Wire), *1-B1(BOR,BOR_01.Out), *1-C1(BAND,BAND_01), *1-C1(TONR,TONR_01.TimerEnable)*

Name	Default	Data Type	Scope
BNOT_01		FBD_BOOLEAN_NOT	MIG_Interface
Usage:	Local Tag		
External Access:	None		
OPC UA Access:	None		

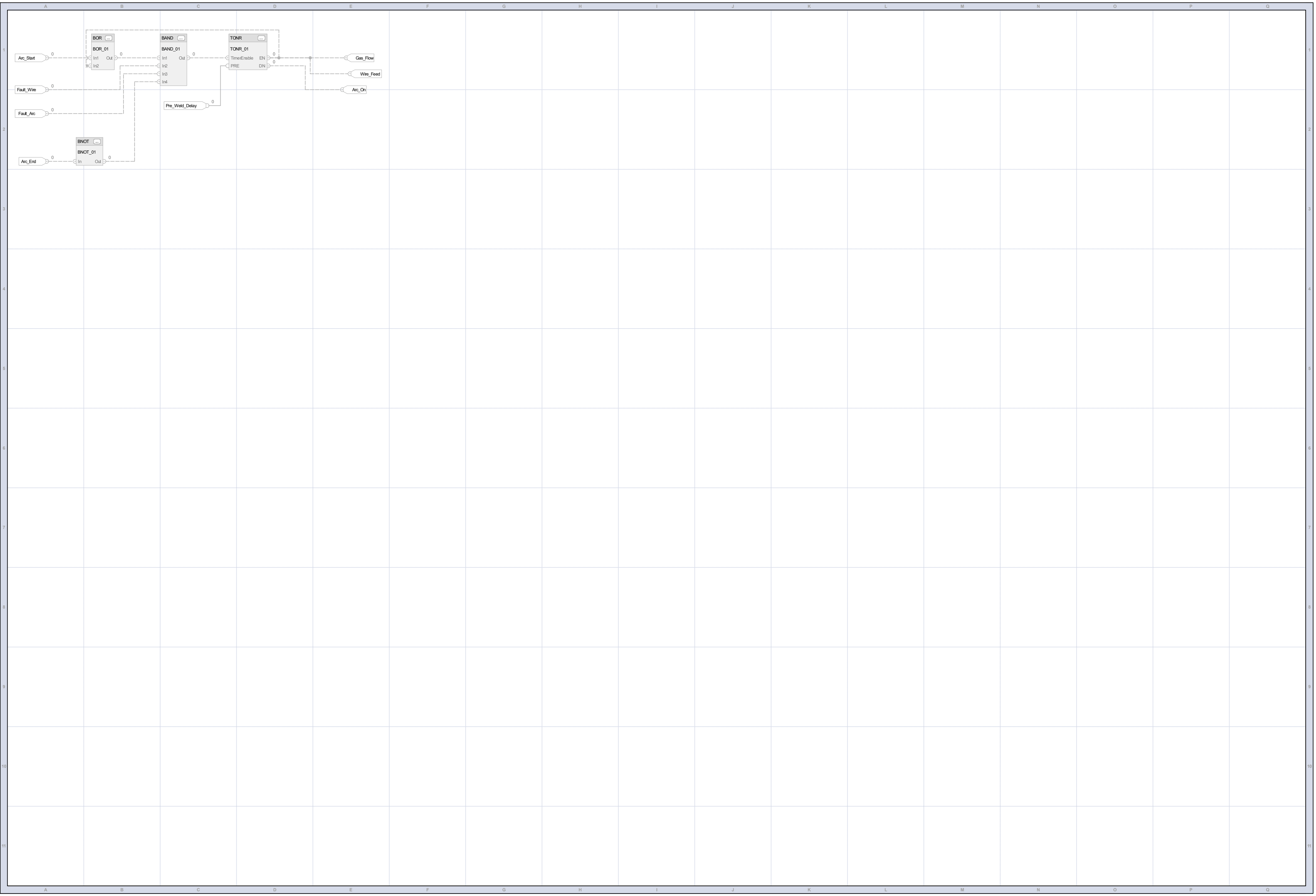
*BNOT_01 - MIG_Interface/Logic - *1-A2(BNOT,BNOT_01), *1-A2(IREF,Arc_End), *1-C1(BAND,BAND_01.In4)*

Name	Default	Data Type	Scope
BOR_01		FBD_BOOLEAN_OR	MIG_Interface
Usage:	Local Tag		
External Access:	None		
OPC UA Access:	None		

*BOR_01 - MIG_Interface/Logic - *1-A1(IREF,Arc_Start), *1-B1(BOR,BOR_01), *1-C1(BAND,BAND_01.In1), *1-C1(TONR,TONR_01.EN)*

Name	Default	Data Type	Scope
TONR_01		FBD_TIMER	MIG_Interface
Usage:	Local Tag		
External Access:	None		
OPC UA Access:	None		

*TONR_01 - MIG_Interface/Logic - *1-B1(BOR,BOR_01.In2), *1-C1(BAND,BAND_01.Out), *1-C1(TONR,TONR_01), *1-C2(IREF,Pre_Weld_Delay), *1-E1(OREF,Gas_Flow), *1-E1(OREF,Wire_Feed), *1-E2(OREF,Arc_On)*


Arc_Start
Fault_Wire
Fault_Arc
Arc_End
BOR
BOR_01
In1
In2
Out
BAND
BAND_01
In1
In2
In3
In4
Out
TONR
TONR_01
TimerEnable
PRE
EN
DN
BNOT
BNOT_01
In
Out
Pre_Weld_Delay
Gas_Flow
Wire_Feed
Arc_On